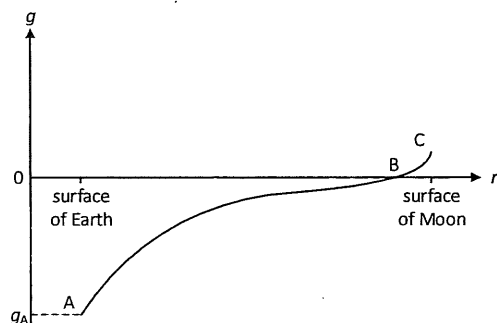


Important note: *denotes questions that may require knowledge from later chapters in order to solve them.

I. MCQs (ACJC – YJC)

1. [ACJC/H2/Prelims 2018/P1/11]

The diagram below shows the variation with distance r between the surface of the Earth A and the surface of the Moon C of the gravitational field strength g . A mass m is at different positions between the Earth and Moon surface.



Which of the following statement is true?

- A At A, mg_A gives the gravitational force due to Earth on m on Earth's surface.
- B At B, the gravitational potential energy of m is maximum.
- C At B, no force acts on mass m .
- D At C, the gravitational potential energy of m is positive.

2. [ACJC/H2/Prelims 2018/P1/12]

Escape velocity from a planet surface is the minimum speed needed for an object at the planet surface to be projected to leave the gravitational influence of the planet.

Given that

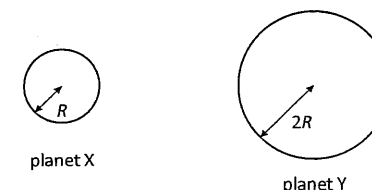
$$\frac{\text{mass of Earth}}{\text{mass of Moon}} = 81; \quad \frac{\text{radius of Earth}}{\text{radius of Moon}} = 3.7$$

What is the ratio of $\frac{\text{escape velocity from surface of Earth}}{\text{escape velocity from surface of Moon}}$?

- A 0.21
- B 4.7
- C 22
- D 480

3. [AJC/H2/Prelims 2018/P1/9]

The diagram shows two planets X and Y, each of them is isolated in space.



Planet X and planet Y have the same mean density. The radius of planet X is half that of planet Y.

What is the ratio of the gravitational field strength at the surface of planet X to the gravitational field strength at the surface of planet Y?

- A 0.5
- B 1.4
- C 2.0
- D 2.8

4. [AJC/H2/Prelims 2018/P1/10]

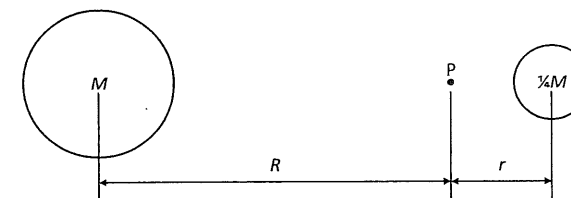
The mass of the Earth is 5.96×10^{24} kg and its mean radius is 6.37×10^6 m.

What is the escape velocity of a body which is at a height 1.00×10^8 m above the Earth's surface?

- A 7.34×10^3 m s⁻¹
- B 1.04×10^4 m s⁻¹
- C 1.11×10^4 m s⁻¹
- D 2.84×10^4 m s⁻¹

5. [DHS/H2/Prelims 2018/P1/10]

Two large masses, one of mass M , the other of mass $\frac{1}{4}M$, are positioned as shown.



A small mass is placed at point P such that it experiences zero gravitational force from the masses.

What is the ratio $\frac{R}{r}$?

- A $\frac{1}{4}$
- B $\frac{1}{2}$
- C 2
- D 4

6. [DHS/H2/Prelims 2018/P1/11]

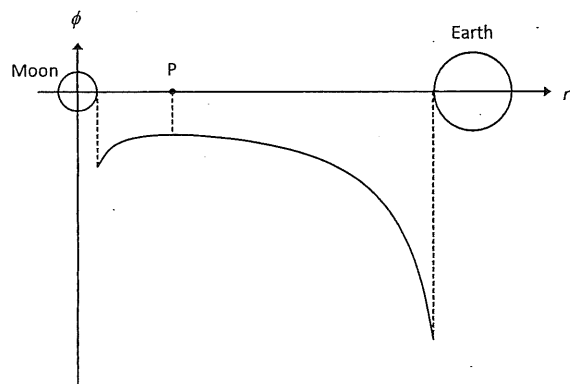
Io and Ganymede are moons of Jupiter. The orbital period of Ganymede is four times that of Io. Io's orbital radius is 4.20×10^8 m.

What is the orbital radius of Ganymede?

- A 1.06×10^9 m
- B 1.68×10^9 m
- C 3.36×10^9 m
- D 2.70×10^{10} m

7. [EJC/H2/Prelims 2018/P1/10]

The gravitational potential ϕ as it varies with distance r from the centre of the Moon to the centre of Earth as shown in the diagram below.



Which of the following statement is **not** true?

- A The gravitational field strength at P is zero.
- B The gravitational field strength at P is the scalar sum of the individual field strengths of the Moon and the Earth at point P.
- C The gravitational field strength at P is the vector sum of the individual field strengths of the Moon and the Earth at point P.
- D The gravitational field strength at P is given by the rate of change of ϕ with respect to r .

8. [HCI/H2/Prelims 2018/P1/12 Modified]

A satellite of mass m is orbiting a planet of mass M at a radius of R .

What is the energy that must be provided to bring the satellite to an orbit of radius $2R$?

- A 0
- B $\frac{GMm}{4R}$
- C $\frac{GMm}{4R}$
- D $\frac{GMm}{2R}$

9. [JC/H2/Prelims 2018/P1/10]

A stationary object is released from a distance $6R$ from the centre of the Earth which has radius R and mass M .

Which one of the following expressions gives the speed of the object on hitting the Earth?

- A $\sqrt{\frac{GM}{R}}$
- B $\sqrt{\frac{GM}{5R}}$
- C $\sqrt{\frac{5GM}{2R}}$
- D $\sqrt{\frac{5GM}{3R}}$

10. [IJC/H2/Prelims 2018/P1/11 Modified]

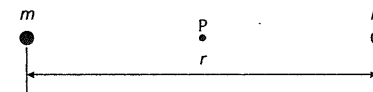
A satellite is orbiting near the Moon's surface. The acceleration at the Moon's surface due to lunar gravity is $\frac{1}{6}g$ and the Moon's radius is $\frac{1}{4}R$ (where g is the acceleration of the Earth and R is the radius of the Earth).

If a satellite orbiting near the Earth's surface has a period of T , what is the period of revolution of the Moon's satellite?

- A $\frac{4}{9}T$
- B $\frac{2}{3}T$
- C $T\sqrt{\frac{2}{3}}$
- D $T\sqrt{\frac{3}{2}}$

11. [JJC/H2/Prelims 2018/P1/9]

The figure below shows two objects of equal mass m separated by a distance r .

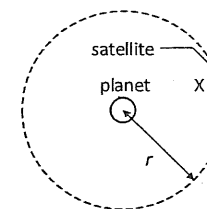


Which of the following gives the correct values of the gravitational field strength and gravitational potential at the mid-point P between the two objects?

	gravitational field strength	gravitational potential
A	$-\frac{8Gm}{r^2}$	$-\frac{4Gm}{r}$
B	$-\frac{8Gm}{r^2}$	0
C	0	$-\frac{4Gm}{r}$
D	0	0

12. [JJC/H2/Prelims 2018/P1/10]

A satellite X is in a circular orbit of radius r about the centre of a spherical planet of mass M .



Which of the following gives the correct expressions for the centripetal acceleration a and the speed v of the satellite?

	centripetal acceleration a	speed v
A	$\frac{Gm}{2r}$	$\sqrt{\frac{Gm}{2r}}$
B	$\frac{Gm}{2r}$	$\sqrt{\frac{Gm}{r}}$
C	$\frac{Gm}{r^2}$	$\sqrt{\frac{Gm}{2r}}$
D	$\frac{Gm}{r^2}$	$\sqrt{\frac{Gm}{r}}$

13. [MI/H2/Prelims 2018/P1/13]

A planet X has half the mass and radius of the Earth. The acceleration due to gravity on the Earth's surface is g . What is the acceleration due to gravity on the surface of planet X?

- A $\frac{1}{4}g$ B $\frac{1}{2}g$ C g D $2g$

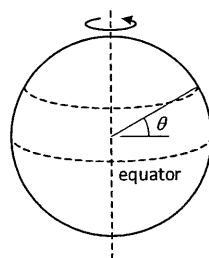
14. [MJC/H2/Prelims 2018/P1/8]

In two widely-separated planetary systems whose suns have masses S_1 and S_2 , planet P_1 of mass M_1 (orbiting sun S_1) and planet P_2 of mass M_2 (orbiting sun S_2) are observed to have circular orbits of equal radii. If P_1 completes an orbit in half the time taken by P_2 , it may be deduced that

- A $S_1 = S_2$ and $M_1 = 0.25M_2$
 B $S_1 = 4S_2$ only
 C $S_1 = 4S_2$ and $M_1 = M_2$
 D $S_1 = 0.25S_2$ only

15. [NJC/H2/Prelims 2018/P1/9]

The Earth rotates on its axis once every 24 hours.



At what latitude θ is the tangential speed of a person one third that of a person living at the equator?

- A 20.5° B 30.5° C 60.5° D 70.5°

16. [NYJC/H2/Prelims 2018/P1/11]

Two satellites X and Y are orbiting at heights of $5R$ and $9R$ above the surface of the Earth respectively, where R is the radius of the Earth.

What is the ratio of their orbital speeds v_X/v_Y ?

- A 1.29 B 1.34 C 1.67 D 1.80

17. [NYJC/H2/Prelims 2018/P1/12]

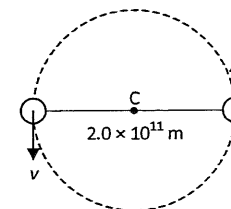
The gravitational potentials on the surface of a planet P and on the surface of its moon Q are -120 MJ kg^{-1} and -20 MJ kg^{-1} respectively.

The minimum amount of energy required to project a 1 kg mass from the surface of Q to the surface of P is E . Which of the following correctly describes the value of E ?

- A $E < 20 \text{ MJ}$
 B $E = 20 \text{ MJ}$
 C $20 \text{ MJ} < E < 100 \text{ MJ}$
 D $100 \text{ MJ} < E < 120 \text{ MJ}$

18. [PJ/C/H2/Prelims 2018/P1/7]

A binary star system consists of two identical stars each of mass $4.0 \times 10^{30} \text{ kg}$ orbiting about their common centre of mass C. The stars are moving with a constant speed v and their centres are separated by a distance of $2.0 \times 10^{11} \text{ m}$.



What is the speed v ?

- A $1.8 \times 10^4 \text{ m s}^{-1}$
 B $2.6 \times 10^4 \text{ m s}^{-1}$
 C $3.7 \times 10^4 \text{ m s}^{-1}$
 D $5.2 \times 10^4 \text{ m s}^{-1}$

19. [RVHS/H2/Prelims 2018/P1/11]

A spacecraft orbiting a planet in uniform circular motion has an antenna detached gently from it. Neglecting air resistance, which statement best describes the motion of this detached antenna?

- A It will move off in a straight line away from the planet.
 B It will take a parabolic path into the planet.
 C It will continue to orbit in uniform circular motion.
 D It will drop straight down into the planet.

20. [RVHS/H2/Prelims 2018/P1/12]

A satellite of mass 80 kg moves from a point where its gravitational potential energy due to Earth is -4800 MJ , to another point where its gravitational potential energy is -1600 MJ .

In which direction does the satellite move and what is the change in gravitational potential between these two points?

- A closer to Earth and a decrease of 40 MJ kg^{-1} of potential.
- B closer to Earth and a decrease of 3200 MJ kg^{-1} of potential.
- C further from Earth and an increase of 40 MJ kg^{-1} of potential.
- D further from Earth and an increase of 3200 MJ kg^{-1} of potential.

21. [SRJC/H2/Prelims 2018/P1/13]

Which of the following is a correct description of a geostationary orbit?

The mass of Earth is $6.0 \times 10^{24} \text{ kg}$.

- A The moon is an example of a geostationary satellite of Earth.
- B A geostationary satellite has an orbital circumference of $2.7 \times 10^8 \text{ m}$.
- C A geostationary satellite moves from North pole to South pole then back to North pole in 24 hours.
- D A geostationary satellite experiences zero net force as it orbits around Earth.

22. [TPJC/H2/Prelims 2018/P1/14]

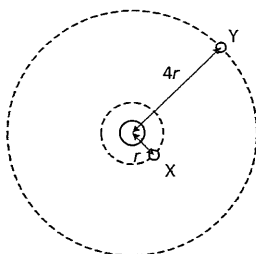
Newton's law of gravitation can be applied to the Earth-Moon system.

Which of the following statements is **not** correct?

- A The value of G at the surface of the Moon is the same as that at the surface of the Earth.
- B The gravitational force between the Earth and the Moon is proportional to the square of the separation of the Earth and the Moon.
- C The gravitational force between the Earth and the Moon is proportional to the mass of the Moon.
- D The orbital period of the Moon about the Earth is independent of the mass of the Moon.

23. [TPJC/H2/Prelims 2018/P1/15 Modified]

Two planets travel around a star in the same direction, in circular orbits. The orbital period of planet X is T . The ratio of the orbital radii is $1/4$.



How many revolutions does planet Y make in the same time T ?

- A 0.125
- B 0.25
- C 0.5
- D 8

24. [VJC/H2/Prelims 2018/P1/8]

Which statement about geostationary orbits is false?

- A A geostationary orbit must be directly above the equator.
- B All satellites in geostationary orbits must have the same mass.
- C The period of a geostationary orbit must be 24 hours.
- D There is only one possible radius for a geostationary orbit.

25. [VJC/H2/Prelims 2018/P1/9]

A satellite of mass 690 kg orbits the Earth along a circular path of radius $7.2 \times 10^6 \text{ m}$. Its orbital speed is 7.5 km s^{-1} . A rocket engine on the satellite is fired. The satellite falls into a lower orbit of radius $6.5 \times 10^6 \text{ m}$ where its orbital speed becomes 7.9 km s^{-1} .

Which statement about the change to the total energy of the satellite caused by the rocket burn is correct? [Mass of the Earth is $6.0 \times 10^{24} \text{ kg}$]

- A The satellite gains approximately $2 \times 10^9 \text{ J}$.
- B The satellite gains approximately $3 \times 10^9 \text{ J}$.
- C The satellite loses approximately $2 \times 10^9 \text{ J}$.
- D The satellite loses approximately $4 \times 10^9 \text{ J}$.

26. [YJC/H2/Prelims 2018/P1/9]

A satellite is at a height h above the surface of the Earth.

If the radius of the Earth is R , and the acceleration due to gravity at the Earth's surface is g , the period of orbit of the satellite will be

- A $2\pi \sqrt{\frac{R+h}{g}}$
- B $2\pi \sqrt{\frac{(R+h)^2}{gR}}$
- C $2\pi \sqrt{\frac{R^2}{g(R+h)}}$
- D $2\pi \sqrt{\frac{(R+h)^3}{gR^2}}$

27. [YJC/H2/Prelims 2018/P1/10]

A planet has two moons with identical mass. The orbital radius of moon 1 and 2 are r and $2r$ respectively.

What is the ratio of $\frac{\text{force exerted by planet on Moon 2}}{\text{force exerted by planet on Moon 1}}$?

- A 4
- B 2
- C 0.5
- D 0.25

II. Structured Questions (ACJC – YJC)

1. [ACJC/H2/Prelims 2018/P3/2]

A satellite of mass 1200 kg is launched from the Earth's surface into a circular orbit at a height of 35 600 km directly above the equator. The radius of the Earth is 6400 km.

Fig. 1.1 shows the variation with distance from Earth's centre r of gravitational potential ϕ due to Earth.

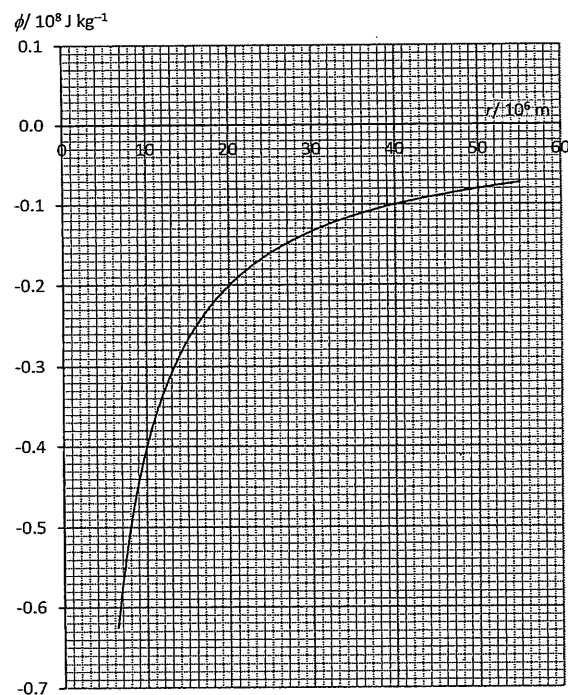


Fig. 1.1

- Explain why the gravitational potential graph is always negative. [2]
- Using Fig. 1.1, show that the gravitational force acting on the satellite by the Earth when it is in orbit is about 270 N. [3]
- Hence or otherwise, calculate the kinetic energy of the satellite when it is at the orbit. [3]
- Use Fig. 1.1 and your answer in c. to determine the minimum amount of energy required to launch the satellite to the orbit. Assume that the kinetic energy of the satellite at the surface of Earth is 0 J. [3]
- The actual energy required to send the satellite to the orbit is different from the energy calculated in d. Suggest a reason why. [1]

[Total: 12]

2. [AJC/H2/Prelims 2018/P3/2]

The Earth may be considered to be a uniform sphere of radius 6.4×10^6 m and its mass is assumed to be concentrated at its centre.

- Show that the mass of the Earth is 6.0×10^{24} kg. [1]
- A satellite of mass 650 kg is to be launched from the Equator and put into geostationary orbit.
 - Determine the radius of the geostationary orbit. [3]
 - Calculate the increase in gravitational potential energy of the satellite during its launch from the Earth's surface to the geostationary orbit. [2]
- Small resistive forces acting on the satellite cause the radius of its orbit to change. Explain why the kinetic energy of the satellite increases due to these resistive forces. [3]

[Total: 9]

3. [CJC/H2/Prelims 2018/P2/2]

In a binary star system, two stars, each of equal mass 3.5×10^{30} kg, rotate about their common centre of mass O which is equidistant from the centres of the stars. The separation between the two centres of the stars is 2.0×10^{11} m.

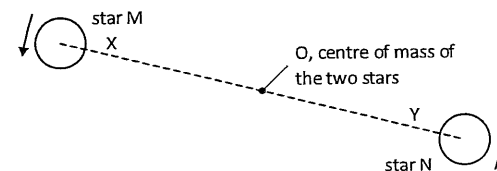


Fig. 3.1

- Define gravitational potential at a point. [1]
- Calculate the gravitational potential at O, the centre of mass of the binary star system. [2]
- An asteroid passes through point O, at a speed v . Determine the minimum speed of the asteroid if it is to escape from the gravitational pull of the binary star system. [3]
- On Fig. 3.2, sketch a graph showing the variation of gravitational potential ϕ along the line XY between the two stars. [2]

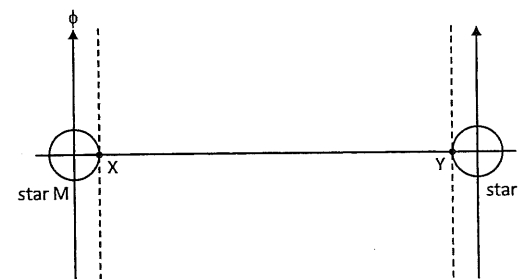


Fig. 3.2

- Hence describe the variation in gravitational potential energy of a mass moving from O towards star M. [2]

[Total: 10]

4. [DHS/H2/Prelims 2018/P2/2 Modified]

Fig. 4.1 shows the variation with distance r from the centre of the Earth of the gravitational field strength g .

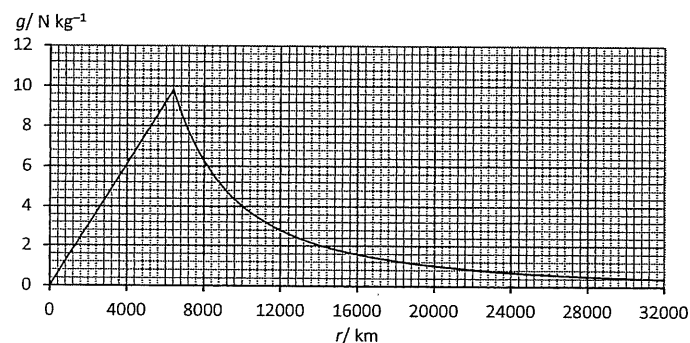


Fig. 4.1

- Use Fig. 4.1 to determine the gravitational force on a man-made satellite of mass 20 000 kg at a distance of 8200 km from the centre of the Earth. [2]
- Calculate the speed of the satellite in a. for it to be orbiting the Earth at constant speed. [2]
- State what is meant by gravitational potential. [1]
 - Use Fig. 4.1 to estimate the gravitational potential at a distance of 10 000 km from the centre of the Earth. [2]

[Total: 7]

5. [HCI/H2/Prelims 2018/P3/8]

- State Newton's law of gravitation. [2]
- Earth rotates about its axis with a period of 24 hours. Assume Earth has a uniform density.
Radius of Earth $R_E = 6.37 \times 10^6$ m
Mass of Earth $M_E = 5.97 \times 10^{24}$ kg
 - Calculate the centripetal acceleration of a man standing at Earth's equator. [2]
 - Hence, calculate the acceleration of free fall at the equator of Earth, and explain why this value may be different from that at the poles. [4]
- The International Space Station (ISS) revolves the Earth in a circular orbit at a height of just 408 km above Earth's surface.
 - Show that the period of the ISS is 1.5 hours. [2]
 - For a circular orbit, the radius, r , and period, T , are related by the relationship

$$T^2 \propto r^3$$

This result is also known as Kepler's Third Law.

A student noted that a point on Earth's equator rotates with a period of 24 hours but the ISS in c.i. orbits with a period of just 1.5 hour.

Comment the apparent discrepancy between the student's observation and Kepler's Third Law. [2]

- An astronaut on the ISS deduced that he must be weightless since he was floating. Comment on his deduction. [2]
- d. *The table shows data relating the Moon orbiting the Earth and an electron orbiting the nucleus of a hydrogen atom.

	Moon orbiting Earth	electron orbiting nucleus
mass / kg	7×10^{22}	9×10^{-31}
speed / m s^{-1}	1×10^3	2×10^7
orbital radius / m	4×10^8	5×10^{-11}

Fig. 5.1

- Using the data from Fig 5.1, determine
 - the de Broglie wavelength of the Moon orbiting the Earth,
 - the de Broglie wavelength of the electron orbiting the nucleus. [3]
- Hence, explain why it is reasonable to treat the Moon in orbit around the Earth as a "particle" but it is not reasonable to treat the electron in orbit around the nucleus as a "particle". [3]

[Total: 20]

6. [JJC/H2/Prelims 2018/P3/2]

- a. State Newton's law of gravitation. [2]
- b. Some of the planets in the Solar System have several moons (satellites) that have circular orbits about the planet. The planet and each of its moons may be considered to be point masses. Show that the radius x of the moon's orbit is related to the period T of the orbit by the expression

$$GM = \frac{4\pi^2 x^3}{T^2}$$

where G is the gravitational constant and M is the mass of the planet. Explain your working. [3]

- c. The planet Neptune has eight moons, each in a circular orbit of radius x and period T . The variation with T^2 of x^3 for some of the moons is shown in Fig. 6.1.

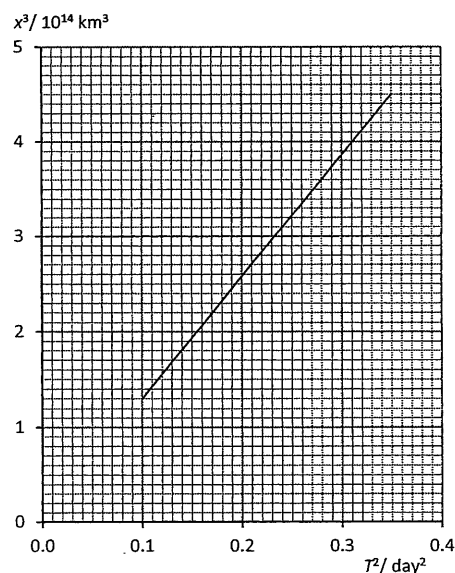


Fig. 6.1

- c. i. Show that the gradient of the line in Fig. 5.1 is $1.7 \times 10^{14} \text{ m}^3 \text{ s}^{-2}$ [1]
 ii. Hence determine the mass of Neptune. [2]
- d. The mass of the planet Uranus is smaller than that of the planet Neptune. [3]
 On Fig. 6.1, sketch the variation with T^2 of x^3 for the moons of Uranus.

[Total: 11]

7. [JJC/H2/Prelims 2018/P3/4]

The variation of the gravitational potential ϕ with distance x from the centre of the Earth is shown in Fig. 7.1. The radius R of the Earth is $6.40 \times 10^3 \text{ km}$.

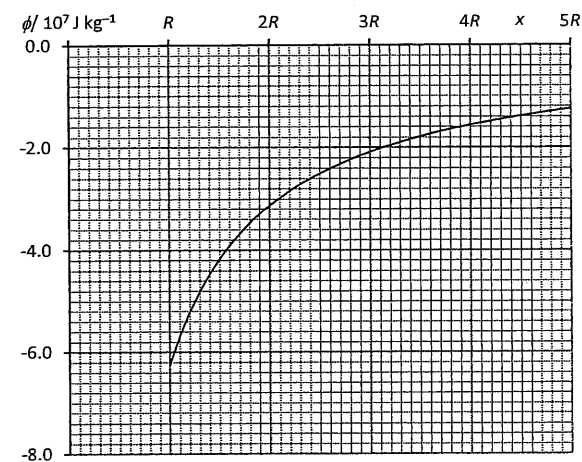


Fig. 7.1

- a. By considering the gravitational potential at the Earth's surface, determine a value for the mass of the Earth. [2]
- b. A meteorite is at rest at infinity. It travels from infinity towards the Earth.
- i. Calculate the speed of the meteorite when it is at a distance of R above the Earth's surface. [3]
 ii. In practice, the Earth is not an isolated sphere because it is orbited by the Moon, as illustrated in 7.2.

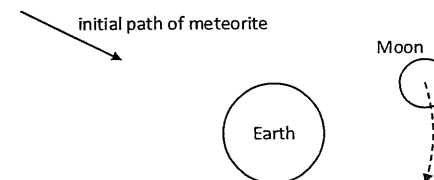


Fig. 7.2

The initial path of the meteorite is also shown.

Suggest two changes to the motion of the meteorite caused by the Moon.

[2]

[Total: 7]

8. [MI/H2/Prelims 2018/P3/2 Modified]

- a. Fig. 8.1 shows the variation of the gravitational potential ϕ with distance r along a line joining the centres of the moon and the Earth.

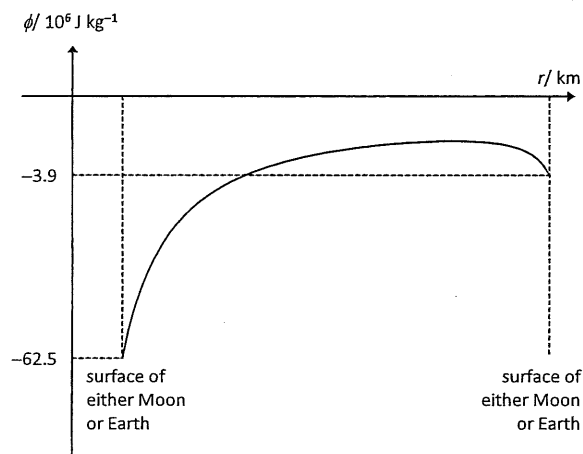


Fig. 8.1

- Explain whether r is measured from the moon or the Earth. [1]
- A 1500 kg meteorite needs a minimum energy of 3.90×10^9 J to reach the neutral point as it travels from the moon to the Earth.
 - Explain what is meant by the neutral point between the moon and the Earth. [1]
 - Determine the gravitational potential at the neutral point. [2]
 - Determine the speed at which the meteorite would hit the Earth. You may neglect atmospheric resistance. [2]
- The mass of the Earth is 5.98×10^{24} kg and the moon takes 27.4 days to orbit the Earth.
 - Show that the distance between the centre of the Earth and the moon is about 384 000 km. [2]
 - Hence, determine the linear speed of the moon. [2]
- An astronaut in a spacecraft orbits around the Earth in a circular path. The astronaut appears to float inside the spacecraft. Students A and B provide some explanations about this observation of weightlessness.
 - Student A claims that the astronaut floats because he has no weight.
 - Student B claims that the astronaut floats because both the astronaut and the spacecraft have the same centripetal force.

Comment on the validity of each of these explanations. [2]

[Total: 12]

9. [MJC/H2/Prelims 2018/P3/8 Part]

- a. A binary star consists of two stars that orbit about a fixed point C, as shown in Fig. 9.1.

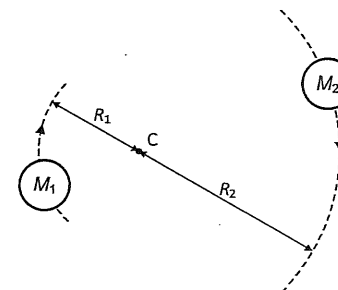


Fig. 9.1

The star of mass M_1 has a circular orbit of radius R_1 and the star of mass M_2 has a circular orbit of radius R_2 . Both stars have the same angular speed ω about C.

- State the formula, in terms of G , M_1 , M_2 , R_1 , R_2 and ω for
 - The gravitational force between the two stars, [1]
 - The centripetal force on the star of mass M_1 . [1]
- The stars orbit each other in a time of 1.26×10^8 s. Calculate the angular speed ω for each star. [1]
- Show that the ratio of the masses of the stars is given by the expression $\frac{M_1}{M_2} = \frac{R_2}{R_1}$. [1]
- The ratio $\frac{M_1}{M_2} = \frac{R_2}{R_1}$ is equal to 3.0 and the separation of the stars is 3.2×10^{11} m.

Determine the radii R_1 and R_2 . [1]
- By considering the expressions in i. and using the data calculated in ii. and iv., determine M_2 . [3]

[Total: 9]

10. [NJC/H2/Prelims 2018/P3/5]

- a. i. State what is meant by a *line of force in a gravitational field*. [1]
 ii. By reference to the lines of force of the gravitational field near the surface of the Earth, explain why the acceleration of free fall near the surface of Earth is approximately constant. [3]
- b. Fig. 10.1 shows some data for Mercury and Pluto.

	mass/ kg	radius/ m	mean distance from Sun/ m
Mercury	3.30×10^{23}	2.44×10^6	57.9×10^9
Pluto	0.131×10^{23}	1.19×10^6	5910×10^9

Fig. 10.1

- i. Show that the escape velocity v of a gas molecule on the surface of Pluto is given by the equation

$$v = \sqrt{\frac{2GM}{r}}$$

where M is the mass of Pluto and r is its radius.

- ii. Calculate the escape velocity of a gas molecule on the surface of Pluto. [2]
 iii. Suggest, using data from the table, why Mercury has no detectable atmosphere while Pluto has a thin atmosphere. [3]

[Total: 10]

11. [NYJC/H2/Prelims 2018/P3/4]

- a. "The magnitude of *gravitational field strength* at a point is proportional to the *gravitational potential gradient* at that point." Explain the terms in *italics*. [2]
- b. Fig. 11.1 shows four equipotential lines at 1000-km intervals in a non-uniform gravitational field.

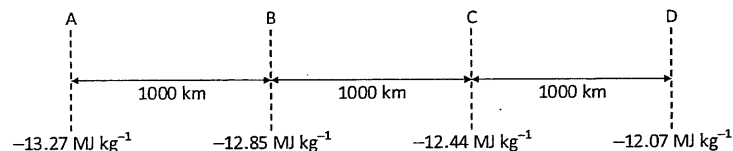


Fig. 11.1

- i. Draw an arrow on Fig. 11.1 to indicate the direction of the gravitational field. [1]
 ii. Calculate the ratio of $\frac{\text{field strength at B}}{\text{field strength at C}}$. [3]
- c. A theoretical long open tube is placed in the gravitational field as shown in Fig. 11.2. A ball is released from rest at B.

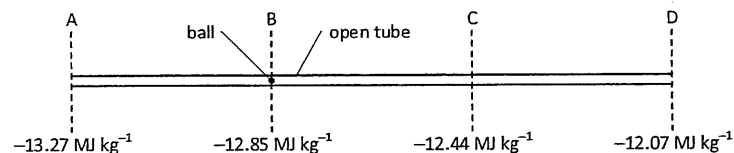


Fig. 11.2

Determine the speed with which the object leaves the tube.

[4]

[Total: 10]

12. [PJC/H2/Prelims 2018/P2/3 Modified]

A space rocket on Earth is fired for a few minutes to provide it with the necessary kinetic energy for it to travel directly to the Moon. Along the line joining the centres of the Earth and Moon, there is a point P where the rocket does not experience any gravitational force. P is 3.46×10^5 km from the centre of the Earth and 3.80×10^4 km from the centre of the Moon as shown in Fig. 12.1.

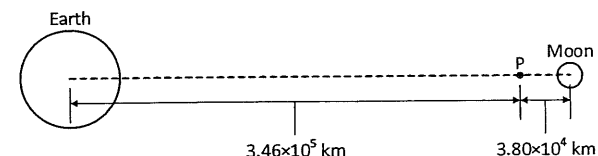


Fig. 12.1

- a. Explain why the rocket does not experience any gravitational force at P. [2]
 b. Suggest two reasons why a return rocket from the Moon would need much less fuel than that required for the outward journey from the Earth. [3]
 c. Given that the mass of the Earth is 5.97×10^{24} kg, calculate the mass of the Moon. [2]
 d. A meteorite of mass 7.5 kg moves towards P under the gravitational attraction of the Earth and Moon. It is initially at rest at a large distance away from P. Calculate the kinetic energy of the meteorite when it reaches P. [3]

[Total: 10]

13. [RVHS/H2/Prelims 2018/P2/3]

- a. i. Define *gravitational potential* at a point in a gravitational field. [1]
 ii. Explain why gravitational potential is always negative. [2]
- b. Fig. 13.1 shows the variation of gravitational potential ϕ between the surface of Moon and the surface of Earth along the line joining their centres.

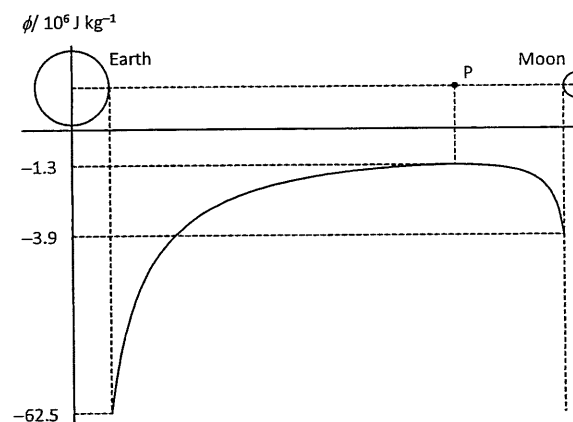


Fig. 13.1

Explain why the gradients of the graph near the surface of Earth and that near the surface of Moon have opposite signs. [2]

- c. Two moons A and B move in circular orbits about a planet, as illustrated in Fig. 13.2.

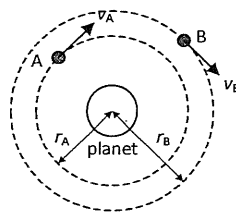


Fig. 13.2

Moon A has an orbital radius $r_A = 1.3 \times 10^8$ m, linear speed v_A and orbital period T_A .

Moon B has an orbital radius $r_B = 2.2 \times 10^{10}$ m, linear speed v_B and orbital period T_B .

- Determine the ratio $\frac{v_A}{v_B}$. [2]
- Show that $\frac{T_A}{T_B} = 4.5 \times 10^{-4}$. [1]
- The planet spins about its own axis with angular speed 1.7×10^{-4} rad s⁻¹. Moon A is always above the same point on the planet's surface. Determine the orbital period T_B of moon B. [2]

[Total: 10]

14. [SRJC/H2/Prelims 2018/P3/8 Modified]

- Define *gravitational field strength*. [1]
 - Derive, from Newton's Law of Gravitation and a.i., the equation for the gravitational field strength of a point mass of mass M_1 , placed in a gravitational field generated by a mass M_2 and at a distance r from M_1 . [2]

- b. Information related to the Earth and the Moon is given below:

$$\frac{\text{Radius of Earth}}{\text{Radius of Moon}} = 3.7$$

$$\frac{\text{Mass of Earth}}{\text{Mass of Moon}} = 81$$

The center-to-center distance of the Moon from the Earth is 3.84×10^8 m and the gravitational field strength due to the Earth at its surface is 9.8 N kg^{-1} .

- Using these data, calculate the gravitational field strength due to the Moon at its surface. [3]
- There is a point on the line between the Earth and the Moon at which their combined gravitational field strength is zero. Calculate the distance between this point and the centre of the Earth. [2]
- The Moon orbits around the Earth with a period of 27.3 days.
 - Calculate the angular speed of the Moon. [1]
 - Calculate the mass of the Earth. [2]
 - Determine the gravitational force between the Earth and the Moon. [2]
 - Tidal action on the Earth's surface causes the radius of the orbit of the Moon to increase by 4.0 cm each year. Using your answer in b.iii.3, determine the change, in one year, of the gravitational potential energy of the Moon. [2]

- Explain, by considering the respective field forces, why gravitational potential energy is negative whereas electric potential energy can be positive or negative. [1]
- The Earth may be assumed to be an isolated sphere of radius 6.4×10^3 km with a mass of 6.0×10^{24} kg. A 2.0 kg mass is projected vertically from the surface of the Earth so that it reaches a maximum altitude of 1.3×10^4 km. Calculate, for this mass,
 - the change in gravitational potential energy, and [2]
 - the speed of projection from the Earth's surface, assuming air resistance is negligible. [2]

[Total: 20]

15. [TPJC/H2/Prelims 2018/P3/3]

Fig. 15.1 shows the variation with distance x of the gravitational potential ϕ between the surface of the Earth and the surface of the Moon. At point P, the gravitational potential is a maximum.

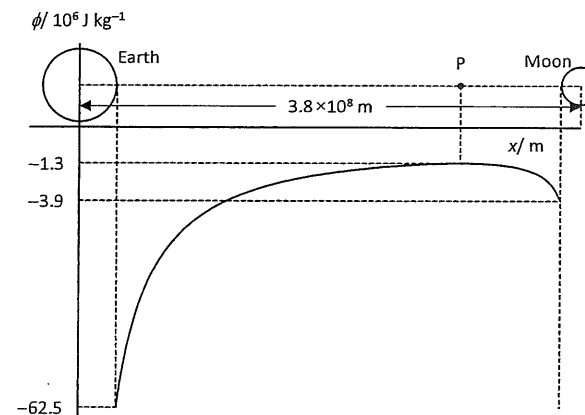


Fig. 15.1

The centres of both masses are separated by a distance of 3.8×10^8 m. The masses of the Earth and the Moon are 6.0×10^{24} kg and 7.4×10^{22} kg respectively.

- Define *gravitational potential* at a point and explain why the values of the potential shown in Fig. 15.1 are negative. [3]
- State how the resultant gravitational field strength at any point between the Earth and the Moon can be deduced from the graph in Fig. 15.1. [1]
 - Determine the distance from the centre of the Earth where the resultant gravitational field strength is zero. [2]
- On Fig. 15.2, sketch a graph to show the variation with distance x of the resultant gravitational field strength g with between the surface of the Earth and the surface of the Moon. Numerical values are not required. [2]

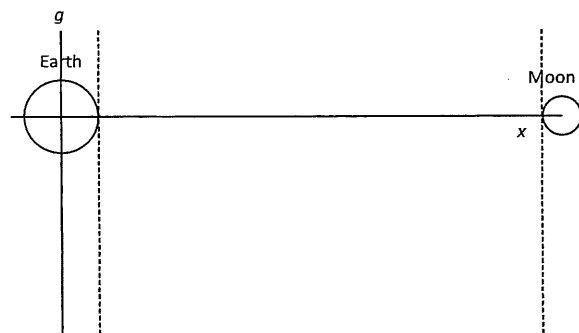


Fig.15.2

- c. A spacecraft is to be launched from the surface of the Earth to reach the surface of the Moon.
Determine the minimum speed required to project the spacecraft from the surface of the Earth. Explain your answers clearly.

[3]

[Total: 11]

16. [VIC/H2/Prelims 2018/P3/8 Part]

- b. Fig. 16.1 shows the dwarf planet, Pluto, and its moon, Charon. These can be considered to be a double planetary system orbiting each other about their joint centre of mass. To simplify the situation, assume that Pluto and Charon have circular orbits of radii r_p and r_c , respectively. The distance between the centres of mass of Pluto and Charon is $r = 1.96 \times 10^4$ km.

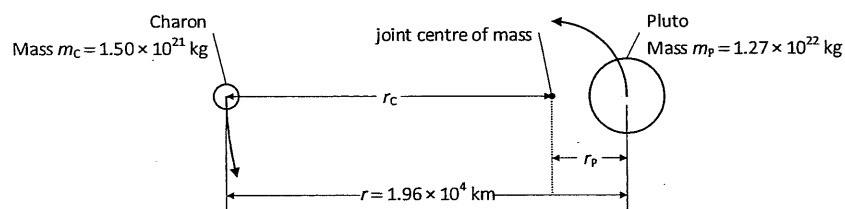


Fig. 16.1

- Explain what is meant by the gravitational field strength at a point. [1]
- Calculate the gravitational field strength at the centre of Charon due to Pluto alone, and state its direction. [3]
- Both Pluto and Charon have the same angular speed about their joint centre of mass.
Show that the ratio of the masses is given by

$$\frac{m_p}{m_c} = \frac{r_c}{r_p}$$

[3]

- Calculate the radius r_c of Charon's orbit. [2]

[2]

- Calculate Charon's orbital period. [2]

[2]

[Total: 11]

17. [YJC/H2/Prelims 2018/P3/3 Modified]

- Define *gravitational potential* at a point. [1]
- Fig. 17.1 shows three equipotential surfaces centred about the Earth, each with its value indicated.

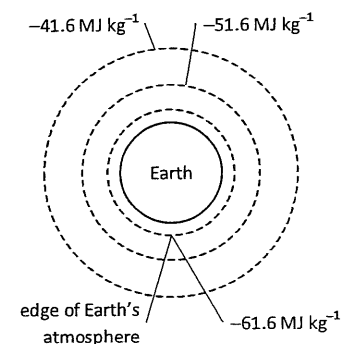


Fig. 17.1

- Explain why
 - the spacing between equipotential surfaces are not the equal, [1]
 - the values of equipotential is negative. [2]
- A rock is released from rest at a point where the potential is -1.0 MJ kg^{-1} .
Calculate its speed when it reaches the edge of the Earth's atmosphere. [3]
- If the radius of the Earth is 6370 km and its mass is $5.97 \times 10^{24} \text{ kg}$, calculate the height of the edge of the atmosphere from the Earth's surface. [3]

[Total: 10]

